The Internet and Higher Education 45 (2020) 100722

Contents lists available at ScienceDirect

The Internet and Higher Education

journal homepage: www.elsevier.com/locate/iheduc

How does the pedagogical design of a technology-enhanced collaborative academic course promote digital literacies, self-regulation, and perceived learning of students?

Ina Blau*, Tamar Shamir-Inbal, Orit Avdiel

Department of Education and Psychology, The Open University of Israel, 1 University Road, P.O.B. 808, Ra'anana 43107, Israel

ARTICLE INFO

Keywords:
Digital literacy skills
Pedagogical design
Online communication and collaboration
Psychological ownership
Self-regulated learning strategies
Cognitive, emotional and social perceived learning

ABSTRACT

The wide expansion of digital technologies in higher education has introduced the need for an examination of the added value of various technological tools for quality teaching and active individual and collaborative learning. The current study explored whether and how the pedagogical design of an academic course, which developed a variety of digital literacy competencies, supported students in regulating collaborative technology-enhanced learning and helped them cope with the sense of psychological ownership over collaborative learning outcomes. In addition, we examined how these issues were expressed in cognitive, emotional and social aspects of students' perceived learning (Caspi & Blau, 2011). During four semesters, we conducted a qualitative analysis on reflective learning diaries, written by 78 graduate students studying education (N = 1870 codes). The bottom-up analysis focused on learning processes that enabled the development of various digital literacies conceptualized by the Digital Literacy Framework (DLF; Eshet-Alkalai, 2012): photo-visual, information, reproduction, branching, social-emotional, and real-time thinking skills. Furthermore, findings highlighted the importance of self-regulation and learning new technologies as an integral part of digital literacies. In addition, social-emotional statements expressed the development of effective communication and collaboration that enable students to cope with a sense of ownership over learning outcomes, and present different levels of teamwork: sharing, cooperation, and collaboration. Qualitative coding provided a more granulated perspective on perceived learning by differentiating between positive and negative aspects of emotional and social retrospection during the learning process. The findings contribute to educational theory by extending DLF and by providing new insights to the literature on students' perceived learning. We discuss the implications for instructional design and adoption of innovative pedagogy in higher education.

1. Introduction

The development of information and communication technology (ICT) has introduced the need for an examination of the added value of various technological tools for quality teaching, as well as for active individual and collaborative learning (Blau & Shamir-Inbal, 2017a; Becker et al., 2017). Technological tools enable instant access to updated digital materials for learners (Akyuz & Yavuz, 2015; Chauhan, 2017). Such technology-enhanced learning enables interactions between students and the content, between students and teachers, and among peers (González Videgaray, 2007; Taşkm & Kandemir, 2010). However, without meaningful integration in teaching-learning processes, digital tools and ubiquitous technologies can be ineffective (Becker et al., 2017). Project-based learning and competency-based learning are examples of the pedagogical trends that aim to create richer, more hands-on experiences for students in academia.

In the following sections we discuss the impact of digital literacy on technology-enhanced learning processes. Following that, we address collaboration, communication and self-regulation in digital learning environments as a part of these literacies. Finally, we discuss how these competencies are reflected in students' perceptions of the learning processes.

1.1. Digital literacy

To conduct effective learning in digital environments, learners are required to develop a set of skills referred to as *digital literacy* (DL; Eshet-Alkalai, 2012). While DL is mostly discussed in the context of education systems (e.g., Ferrari, 2012; Porat, Blau, & Barak, 2018), the Horizon report for higher education (Alexander et al., 2019; Becker

* Corresponding author.
E-mail addresses: inabl@openu.ac.il (I. Blau), tamaris@openu.ac.il (T. Shamir-Inbal), oritavdiel@walla.co.il (O. Avdiel).

https://doi.org/10.1016/j.iheduc.2019.100722
Received 26 June 2018; Received in revised form 1 May 2019; Accepted 8 December 2019
Available online 13 December 2019
1096-7516/ © 2020 Elsevier Inc. All rights reserved.